

Anita's Veg Café

Business Challenges & Insights

Turning receipts into revenue insights

Café Analytics Case Study

Understanding customer value, menu preferences, loyalty impact, rewards, and churn through SQL.

Data Analysis • Restaurant Case • SQL Mastery

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Contents

0.1 Project Objective	1
1 Core Business Challenges	1
2 Data Model	1
3 Analytical Missions	2
3.1 Mission 01 — Spending & Visits	2
3.2 Mission 02 — Menu Preferences	2
3.3 Mission 03 — Loyalty Impact	2
3.4 Mission 04 — Rewards & Churn Alert	2
4 SQL Toolkit	2
5 Growth Dashboard	3
6 Learning Outcomes	3
7 Conclusion	3

0.1 Project Objective

Anita's Veg Café is a business intelligence and SQL case study that shows how everyday restaurant transactions can be transformed into practical growth strategies. The café begins as a home-kitchen venture serving simple, affordable vegetarian food and grows into a data-driven business investigation. The objective is to understand what drives café success by analyzing customer spending, visit behavior, menu choices, loyalty-program activity, reward points, and possible customer churn.

Data turns a simple receipt into a decision about revenue, retention, and growth.

1 Core Business Challenges

01 Customer Value

Revenue Growth

Who are the most valuable customers? How much do they spend, and how often do they visit? The answers support VIP promotions and retention campaigns.

02 Menu Preferences

Menu Optimization

Which dishes attract customers and keep them coming back? These insights help Anita focus on best-sellers, reduce waste, and plan complementary dishes.

03 Loyalty Impact

Customer Retention

Does loyalty membership change customer behavior? Before-and-after analysis reveals whether the program increases engagement.

04 Reward & Churn

Smart Growth

How should rewards be calculated, and which customers may be going inactive? The project combines points logic with a 30-day churn alert.

2 Data Model

The database uses three connected relational tables. The sales table is the central transaction table, linking customer activity to menu information and loyalty membership.

Table	Purpose	Key Columns
sales	Every customer order	customer_id, order_date, product_id
menu	Product names and prices	product_id, product_name, price
loyal_customers	Loyalty membership dates	customer_id, join_date

3 Analytical Missions

3.1 Mission 01 — Spending & Visits

Query	Question	Business Value
Q1	Total amount spent by each customer	Identifies high-value customers.
Q2	Distinct days on which each customer ordered	Measures visit frequency and engagement.

3.2 Mission 02 — Menu Preferences

Query	Question	Business Value
Q3	First dish ordered by each customer	Shows the product that created the first impression.
Q4	Most popular dish overall	Reveals the café's star menu item.
Q5	Most frequently ordered dish per customer	Shows individual repeat preferences.

3.3 Mission 03 — Loyalty Impact

Query	Question	Business Value
Q6	First dish ordered after joining loyalty	Measures the first post-enrollment action.
Q7	Last order before joining loyalty	Captures pre-loyalty behavior.
Q8	Pre-loyalty item count and spending	Creates a baseline for comparison.

3.4 Mission 04 — Rewards & Churn Alert

Query	Question	Business Value
Q9	Points earned by each customer	Converts spending into reward currency.
Q10	Points earned during the first loyalty week	Tests the time-limited double-points offer.
Q11	Customers with no order in the last 30 days	Flags possible churn before customers are lost.

4 SQL Toolkit

JOINS Connect sales, menu, and loyalty tables.

SUM / COUNT Calculate spending, items, and order totals.

ROW_NUMBER / RANK Find first, last, and most frequent products.

CASE WHEN Apply product-specific and tiered reward rules.

DATEDIFF Calculate loyalty windows and date ranges.

MAX + Filter Detect customers whose latest order is too old.

5 Growth Dashboard

Targeted Promotions

Identify VIP customers and personalize offers for higher revenue and stronger retention.

High revenue impact

Menu Optimization

Focus on best-sellers, reduce waste, and introduce dishes that complement popular choices.

Medium-high efficiency impact

Loyalty Enhancement

Measure program ROI and refine the experience to increase member engagement.

High retention impact

Smart Rewards

Design points that encourage frequent visits and selected purchases without weakening profitability.

Medium-high profit impact

6 Learning Outcomes

This project demonstrates that meaningful business intelligence does not require millions of rows. A small, well-structured dataset can reveal customer and product patterns when the right questions are asked.

- Relational data modeling
- Aggregations and grouping
- Loyalty and behavioral analytics
- Date-based analysis
- Business translation skills
- Multi-table SQL joins
- Window functions
- Conditional reward logic
- Churn-risk identification
- Real-world CRM patterns

7 Conclusion

Anita's Veg Café is more than a restaurant sales analysis. It is an end-to-end business intelligence case study connecting customer value, menu performance, loyalty strategy, reward systems, and retention planning.

By analyzing 11 business questions across four missions, the project shows how raw sales data can support decisions about revenue growth, menu optimization, customer retention, smart rewards, and early churn prevention.

Anita's Veg Café

Where homely flavors meet data-driven insight.